The effects of prenatal infection on the ultrasonic vocalisations of Sprague-Dawley pups

PSYC 489 Research Topic, Victoria University of Wellington

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Wordcount: 6953

Acknowledgements

Thank you to Bart Ellenbroek for his supervision, to Joyce Colussi-Mas for animal handling training, to Mahima Yapa Nanayakkara for animal husbandry and to Meyrick Kidwell for his guidance on the project.

Abstract

A robust link between prenatal infections and ensuing neurodevelopmental abnormalities, including ASD, schizophrenia, and mood disorders, has been well-established. Ultrasonic vocalisations, specifically separation calls of ~40kHz, may reliably index a neonatal rat pups' developmental state and provide an early marker of neurodevelopmental abnormalities. The present study sought to investigate a dose-dependent effect of the administration of the viral mimic polyinosinic-polycytidylic acid (poly (I:C)) into pregnant Sprague-Dawley dams on alterations in the acoustic parameters of the USVs of neonatal pups. Pregnant dams were subcutaneously injected with either saline, 2mg/kg poly(I:C) or 5mg/kg poly(I:C) on GD12. Total call number, total and mean call duration, mean bandwidth, and percentage of harmonic syllables were measured and analysed from PND7 old pups. No significant main effect of sex or dose, and no significant interaction effect (sex*dose) found for across any of the dependent variables ($p>0.05$). Notwithstanding, our exploratory data warrant further investigation with a sample of a larger size.

Appropriate neurodevelopment is a complex and highly orchestrated process shaped by a mixture of genetic and environmental factors and their interactions. However, we are especially vulnerable to environmental insults during the prenatal period, a time characterised by the rapid establishment of key brain processes and networks (Han et al., 2021). Consequently, any disturbances during these critical periods can alter the typical developmental trajectory and lead to lasting effects for the individual. Following the 1964 rubella pandemic, a subsequent rise in the incidence of autism spectrum disorder (ASD) and schizophrenia (SZ) was observed, from 1% to 13% and 20%, respectively (Prins et al., 2018). Since, many historic outbreaks of influenza, measles and polio have confirmed associations with ASD, SZ and mood disorders (Al-Haddad et al., 2019; Boksa, 2008; Mednick et al., 1988). There now is ample evidence highlighting the importance of maternal and fetal immune systems in mediating neural development, suggesting maternal immune activation during the prenatal period leads to increased risk in the offspring for developing neuropsychiatric abnormalities later in life, including ASD (Atladóttir et al., 2010), SZ (Khandaker et al., 2013) and major depressive disorder (MDD) (Al-Haddad et al., 2019).

The exact mechanisms underpinning the relationship between maternal infection and an increased risk for neurodevelopmental disorders in offspring remain little understood (Goeden et al., 2016). The infectious agent itself is unable to cross the blood-placenta barrier, which decreases the likelihood of it being a direct causal factor (Robbins & Bakardjiev, 2012). A more plausible explanation is the maternal immune activation (MIA) hypothesis, which posits that the infectious agent activates the immune system of the mother, causing an increase in proinflammatory cytokines, notably interleukin-6 (IL-6) (Patterson, 2011). These cytokines can cross the blood-placenta barrier and activate microglia within the fetus (Han et al., 2021). Microglia have an essential role in synaptic pruning and carving of functional neural circuits, so activation can subsequently lead to perturbations in neuronal development (Han et al., 2021; Prins et al., 2018). Support for this hypothesis originates from the abundance of studies cited throughout this paper demonstrating MIA via an immunostimulant leads to unfavourable behavioural outcomes in offspring. Additionally, children of mothers with inflammatory conditions, such as autoimmune conditions or obesity, have increased risk for later neurodevelopmental abnormalities (Han et al., 2021).

Two findings threaten the MIA hypothesis. Firstly, behavioural outcomes are independent of the type of infection (bacterial, viral or fungal), despite each inducing varying inflammatory responses (Jouda et al., 2019; Tioleco et al., 2021). Secondly, the severity of the infection is also independent of the outcome, with even mild infections (such as UTIs) increasing the risk (Al-Haddad et al., 2019). Strikingly, mild infections do not increase proinflammatory cytokines. Goeden et al. (2016) found that a moderate immune challenge following a 2mg/kg dose of the immunostimulant polyinosinic:polycytidylic acid (poly(I:C)) disrupted fetal neurodevelopment, despite no observed increase in placental proinflammatory cytokine levels. However, they did find an upregulation of placental tryptophan and subsequent conversion to 5-HT, which lead them to hypothesise that perhaps a disruption in 5-HT-dependent neurogenic processes during this critical neurodevelopment period precipitates unfavourable functional outcomes. The maternal serotonin hypothesis (MSA) thus posits that serotonin assumes an integral structural role in early development and modulates critical neurodevelopmental processes, therefore increases in placental serotonin may lead to increased risk for neurodevelopmental abnormalities later in life (Harrington et al., 2013; Goeden et al., 2016).

Despite the outstanding debate surrounding the mechanisms, it remains of particular importance, given the current COVID-19 pandemic, to elucidate the mechanisms and risk factors involved in neurodevelopmental disorders. The evidence strongly suggests that pregnant women and their fetuses are at high risk for developing enduring complications. It is therefore imperative to understand the causes and detect these abnormalities early to provide prompt intervention.

Ultrasonic vocalisations (USVs), which succeed the human threshold for hearing (of 20kHz), serve an important communicative function across the lifespan of the rodent and are a well-known indicator of early functional development in neonatal pups (Jouda et al., 2019). Neonatal rat pups predominantly emit USVs around 40kHz, ranging from 30 to 65 kHz (Brudzynski et al., 1999), typically when separated from their mothers and littermates. Playback experiments of these 40kHz calls, but not playback of a 40kHz tone or white noise, have been shown to trigger retrieval behaviour in mothers, which supports the affiliative function of these calls as socio-affective isolation calls, purposed to communicate distress to the mother and stimulate retrieval (Wöhr & Schwarting, 2008). Neonatal pups given anxiolytic drugs reliably show a reduction in the number of these isolation calls (Fish et al., 2004; Hodgson et al., 2008), which implies these calls reflect negative affective states such as stress and anxiety. These isolation calls follow a typical developmental trajectory of an inverted-U profile, with calls peaking around PND10 and then declining with age (Stark et al., 2020). Typically with age, the frequency of these calls becomes higher, the duration becomes shorter, and the calls gain complexity (Stark et al., 2020). It is believed this pattern reflects pups gaining independence from their mother; as pups age, their vision, hearing and locomotion improves, so they become much less reliant on their mother and more resilient to stress (Stark et al., 2020). The socio-affective function of these calls has relevance for the validity of animal models, as many neurodevelopmental disorders feature shared deficits in social functioning and communication (Jouda et al., 2019). Especially regarding the current challenges in creating sensitive behavioural assessments for social behaviour and communication in animals, USVs provide a reliable readout of an animal's behaviour which we can use to evaluate potential neuropsychiatric abnormalities (Jouda et al., 2019).

The maternal immune activation (MIA) model is a widely used paradigm which allows us to stimulate the immune system of the mother to evaluate any resulting functional and behavioural changes in the offspring (Jouda et al., 2019). Most often, lipopolysaccharide (LPS) is used to simulate a bacterial infection, and poly(I:C) is used to simulate a viral infection. Both methods produce qualitatively and quantitatively different effects on cytokine production, t-cell proliferation, types of cells and inflammatory mediators recruited, which might account for differences in functional outcomes observed across studies (Jouda et al., 2019). However, many studies have also noticed differences in functional outcomes depending on the timing of the infection or the dosage (Jouda et al., 2019). In the present study, we sought to investigate the functional outcomes following an immune challenge from both a low (2mg/kg) and a high (5mg/kg) poly(I:C) dose, which is of paramount relevance given COVID-19 is a viral infection. As the MIA hypothesis would predict, maternal infection will lead to neurodevelopmental challenges in the offspring, and as the MSA hypothesis would predict, even low doses (2mg/kg) should disrupt fetal neurodevelopment (Goeden et al., 2016). Therefore, we might expect to see measurable alterations in USVs following both poly(I:C) doses.

Few studies have yet investigated the effects of maternal poly(I:C) administration USV emissions in rat pups, likely due to the prior dominance of transgenic mouse models (Potasiewicz et al., 2020). In the present study, we have chosen to investigate the effects in rat pups as they boast a richer acoustic communication system than mice, making them an appropriate model to study socio-communicative deficits in particular (Caruso et al., 2020). Despite limited research, the majority of studies investigating the consequences of prenatal poly(I:C) treatment specifically in rat pups have consistently found that prenatal poly(I:C) treatment reduces the total number of isolation calls in offspring as compared to controls (Chou et al., 2015; Potasiewicz et al., 2020; Piotrowska et al., 2019). Given that neurodevelopmental disorders are characterised by deficits in social functioning and communication (Hommer & Swedo, 2015), this gives the MIA model good face validity. In line with these findings, we might expect a similar reduction in offspring call rate following prenatal poly(I:C) treatment.

Duration and bandwidth (the range of frequencies spanned in a single call) are two acoustic characteristics found to vary across the lifespan of the healthy rat pup. Typically as rats age, the duration of these isolation calls will decrease, while complexity increases, presumably reflecting the independence gained from their mother (Stark et al., 2020). Little is known about the duration of these calls, but bandwidth reflects the complexity of a call (Brudzynski et al., 1999). Research documenting changes in duration or bandwidth following prenatal poly(I:C) treatment has been inconsistent. Potasiewicz et al. (2020) found an effect of poly(I:C) treatment on call rate in Sprague-Dawley rat pups, no significant effects on call duration or bandwidth were observed. Additionally, Wilkin-Krug et al. (2022) also reported no effect of poly(I:C) on duration, but in contrast to Potasiewicz et al. (2020), they found that treatment decreased the bandwidth of calls. Piotrowska et al. (2019) also reported a reduction in bandwidth following poly(I:C) treatment.

Harmonic syllables are another acoustic feature of USVs, consisting of one main call element and an additional call fragment of a different frequency above the main call (Caruso et al., 2020). Despite the specific meaning of this call category remaining largely unknown (Premoli et al., 2021), human infant cries, which bear some similarities to harmonic syllables, are understood to reflect the emotional state of the baby (Malkova et al., 2012). Additionally, following a painful experience, the harmonic segments of a human infant cry are perceived as more urgent in their environment, stimulating maternal care (Cox et al., 2012; Porter et al., 1986). Thus, isolation calls are generally understood to index a rat pup's emotionality. Despite a paucity of studies in rats, studies in mice have revealed that maternal poly(I:C) administration reduces the number of harmonics in offspring (Premoli et al., 2021), particularly for males (Malkova et al., 2012).

Sex differences in neurodevelopmental disorders are well described. Males are disproportionately affected, two to four times more likely than females to be diagnosed with a neurodevelopmental disorder (Han et al., 2021). The reasons remain largely unclear, but this disparity could reflect gender-biased instruments or the higher presentation of internalising behaviour in females, leaving them underdiagnosed (Han et al., 2021). However, research has shown that male brain has a more inflammatory environment than the female brain during development, resulting in male vulnerability to MIA and potentially accounting for the observed discrepancies (Han et al., 2021). The literature surrounding sex differences in USV emissions following poly(I:C) treatment is inconsistent. Wilkin-Krug et al. (2022) report that following poly(I:C) administration, sex did not affect call number or bandwidth, but male rat pups did produce longer calls than females, while Potasiewicz et al. (2020) found male rat pups following prenatal poly(I:C) treatment emitted significantly more calls than females. In contrast, Piotrowska et al. (2019) found bandwidth was reduced for both females and males pups following poly(I:C) treatment, but a greater reduction in call rate was found in male pups. Such inconsistencies warrant further research.

The present study sought to explore if poly(I:C) treatment leads to behavioural outcomes consistent with neurodevelopment disorders and if these outcomes depend on dose or sex. We firstly hypothesise that poly(I:C) at both a low (2mg/kg) and high (5mg/kg) dose will reduce the total number of calls compared to saline controls. This corroborates research indicating that the severity of infection is independent of the behavioural outcome (Al-Haddad et al., 2019) and that even a low dose of 2mg/kg poly(I:C) is sufficient to produce abnormal behavioural outcomes (Goeden et al., 2016). Finding a reduction in call rate following a low dose of 2mg/kg of poly(I:C) would advance the MSA hypothesis as a possible underlying mechanism. Such a finding would also align with the majority of research that has already found reductions in call rate in rat pups following maternal poly(I:C) treatment (Chou et al., 2015; Potasiewicz et al., 2020; Piotrowska et al., 2019).

Further, we will investigate dose- and sex-dependent differences in three acoustic characteristics of these calls, 1) bandwidth, 2) duration and 3) harmonic syllables. The majority of research investigating changes in USV duration following poly(I:C) treatment has found no effects (Wilkin-Krug et al., 2022; Potasiewicz et al., 2020). In line with this, we do not expect to see an effect across treatments on call duration. Regarding changes in call bandwidth, the literature is contradictory (Potasiewicz et al., 2020; Wilkin-Krug et al., 2022; Piotrowska et al., 2019). As bandwidth is understood to reflect the complexity of a call, we predict, in line with Wilkin-Krug et al. (2022) and Piotrowska et al. (2019), that bandwidth will decrease following maternal immune activation. Harmonic syllables are understood to reflect the emotionality of a call and considering neurodevelopmental disorders are characterised by deficits in emotional rapport and expression (Hommer & Swedo, 2015), it is reasonable to expect in line with findings from Premoli et al. (2021) and Malkova et al. (2012) that poly(I:C) treatment will reduce the number of harmonic syllables.

Finally, we will examine sex differences following prenatal poly(I:C) treatment. Despite inconsistencies within the literature, it appears male offspring are disproportionately affected by prenatal poly(I:C) treatment (Potasiewicz et al., 2020; Piotrowska et al., 2019; Wilkin-Krug et al., 2022). Considering the human evidence of increased prevalence of neurodevelopmental disorders in males compared to females, and males experiencing a comparatively more inflammatory environment during neurodevelopment (Han et al., 2021), it is reasonable to expect greater alterations in the outcome parameters of USVs in male pups following prenatal poly(I:C) treatment.

In the present study, pregnant Sprague-Dawley dams were injected subcutaneously with either saline, 2mg/kg of poly(I:C) or 5mg/kg of poly(I:C) on GD12 to investigate a dose-dependent effect on baseline USV emission in neonatal pups. Data on heart rate variability and USVs were collected on PND7 and PND14 from one male and one female per litter. Heart rate variability data were collected for another project and will not be discussed further in this paper. Due to unforeseen technical issues, USV recordings from PND14 recordings were unviable and unable to be used for analysis in this study. Therefore, call rate, mean and total call duration, mean bandwidth and percentage of harmonic syllables from PND7 USV data will be analysed across dosage and sex.

Method

Subjects

Across the two separate cohorts used in the experiments a total of 19 Sprague-Dawley dams were used and 38 pups were recorded. Exactly half of the pups recorded were male. Each cohort was divided into three treatment groups: saline, 2mg/kg poly(I:C) and 5mg/kg poly(I:C). The first cohort used in our pilot study consisted of one saline litter, two 2mg/kg poly(I:C) litters and two 5mg/kg poly(I:C) litters. In our second cohort, there were five saline litters, four 2mg/kg poly(I:C) litters and five 5mg/kg poly(I:C) litters. All rats were housed in individually ventilated cages on a rotational rack. Rats were kept on a reverse day/night cycle, where lights were turned off at 7am, and turned back on at 7pm. The facility was kept at a constant temperature of 21°C (±2 °C). Animal husbandry included the cleaning of cages once a week, and food and water was available *ad libitum*.

Procedure

For the mating, each female rat was paired with a male. Pairs were placed together in a cage lined with a grate on the bottom, which allowed us to inspect for vaginal plugs. Following the pairing, cages were checked twice a day for vaginal plugs. The presence of a vaginal plug was deemed gestational day (GD)1, after which the pair were then separated and rehoused in individual cages.

On GD12, pregnant dams were randomly assigned to one of the three treatment groups, and were subcutaneously injected with either saline (1ml/kg), 2mg/kg poly(I:C) or 5mg/kg poly(I:C). Subcutaneous injection was chosen as it is the least stressful application route (Wilkin-Krug et al., 2022). Body weight of dams were recorded immediately before and 24 hours following the subcutaneous injection, as previous research has shown that the increase in bodyweight over a 24h period is significantly less in poly(I:C) treated dams compared to saline treated controls (Wilkin-Krug et al., 2022). Additionally, temperature was recorded using a non-contact, infrared thermometer (Pro’s Kit, MT-4612) immediately before and 2, 4 and 24 hours following the injection. Previous research has shown that poly(I:C) can lead to small increases in temperature up to 4 hours following administration (Wilkin-Krug et al., 2022).

Pregnant dams were left undisturbed until GD17, after which the cages of dams were filled with nesting material and inspected twice daily for the birth of pups. Dams were checked from the outside of the cage to ensure minimal stress for the animal. The presence of pups was deemed PND1.

On PND7, one male and one female pup were randomly selected from each litter to be recorded. Baseline heart rate variability (HRV) data, as well as USVs were recorded for approximately 8-10minutes. Each pup was placed into a temperature controlled ECGenie holder which was turned on 30 minutes prior to recording in order to allow the holder to heat up. Temperature control is critical because previous research has shown that USVs can be influenced by temperature (Stark et al., 2020). The ECGenie holder was secured by two bands to minimise excessive movement or pups escaping the holder. ECG data was recorded using ECGenie recording software and USV audio data was collected using an Ultramic UM250K and the recording software Audacity. After recording, each pup was returned back to their mother and litter mates. Male pups were marked ‘I’ and female pups were marked ‘II’ using a permanent marker, to ensure the same pups were used for the PND14 recordings. Pups were remarked each subsequent day to keep track of the pups we were measuring. Importantly, pups were separated for no longer than 15 minutes from their mother during recordings, as previous research has indicated that separation lasting more than 15 minutes can increase maternal pup-licking behaviour (Liu et al., 1997), which risks the marker rubbing off and losing track of the pup for PND14 measurements.

On PND14, ECG and USV data were recorded again exactly as described above. Pups infrequently escaped the apparatus, in which case we stopped the recording and started a new one, while ensuring not to keep the pup separated for over 15 minutes. Despite our best efforts many pups still were quite active (particularly at PND14, when they didn't fit as well into the ECGenie holder) which likely contributed to some of the noise contaminating our data.

Following completion of the PND14 recordings, all rat pups from each litter were euthanised via CO_2 asphyxiation and cervical dislocation.

Data and statistical analysis

USV audio files recorded using Audacity were prepared for further analysis using Raven Pro (version 1.6.3). The parameters plugged into the software to locate calls were between 30kHz and 60kHz, between 20ms and 2000ms, and no more than 30ms between each call. Any calls exceeding these parameters were not detected nor analysed. Within these parameters, the software detected the minimum and maximum frequency of each call, the average frequency of each call and the duration of each call (denoted from begin to end time).

This is a software specifically designed for the analysis of bird calls, so it is not fully equipped to detect USVs. As a result, the quality of the audio files affected the detection accuracy of the software. Additionally, there was an unexplained consistent frequency band at around 45kHz that interfered with the software's ability to accurately detect the frequency range of each call, and many of the audio files were contaminated with noise throughout, which was often incorrectly captured by the software as a USV. As a result, each USV was subsequently manually analysed, to ensure correct frequency ranges were obtained, to ensure no calls were missed, and to ensure no noise was mistakenly recorded as a call. Additionally, only the first 8 minutes of each recording was analysed to allow for comparison between recordings. The software was also unable to detect harmonic syllables, so these were also detected and analysed manually. Harmonic syllables were included in the analysis if they were double the frequency above the main call and following the same sonographic structure (*Fig. 1.*). Infrequently, some harmonic calls overlapped the main call, and these were not included in the analysis (*Fig. 2.*).

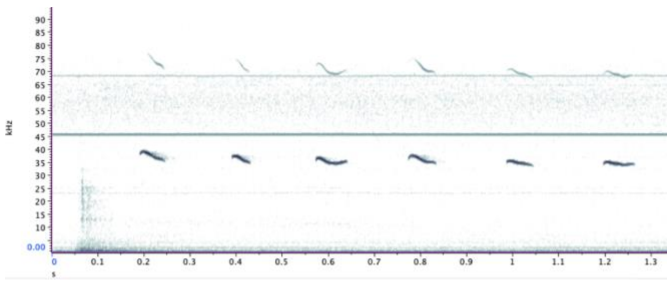


Fig. 1. An example of a series of USVs with harmonic syllables meeting inclusion criterion for the harmonic analysis

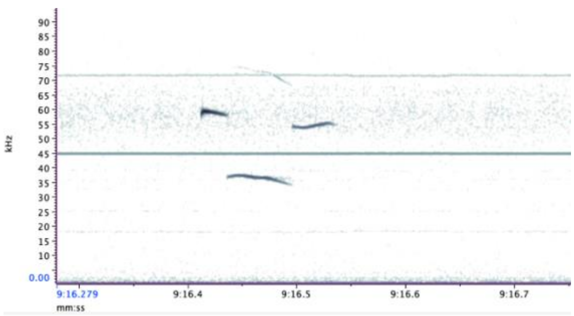


Fig. 2. An example of a USV that was not considered a harmonic syllable

Statistical analysis was conducted using Jamovi Statistical Software. The dependent variables were total number of calls, total call duration, mean call duration, mean call bandwidth and percentage of harmonic syllables (as a percentage of total call number). The independent variables were sex (male/female) and dosage (saline/2mg/kg poly(I:C)/5mg/kg poly(I:C)). Two-way ANOVAs were conducted to analyse main effects for sex and dosage, as well as an interaction effect between sex and dosage.

Results

One male neonate in the 2mg/kg poly(I:C) condition was excluded from the entire analysis due to unviable data from excessive noise contaminating the recording. Data presented in Fig. 3. is the combined PND7 data from both cohort 1 and cohort 2. Data presented in Fig. 4. is limited to cohort 2 data. No analysis or comparison was conducted with the PND14 data as all recordings were unviable due to excessive noise. For complete ANOVA tables of the combined cohort 1 and cohort 2 data, see Appendix A.

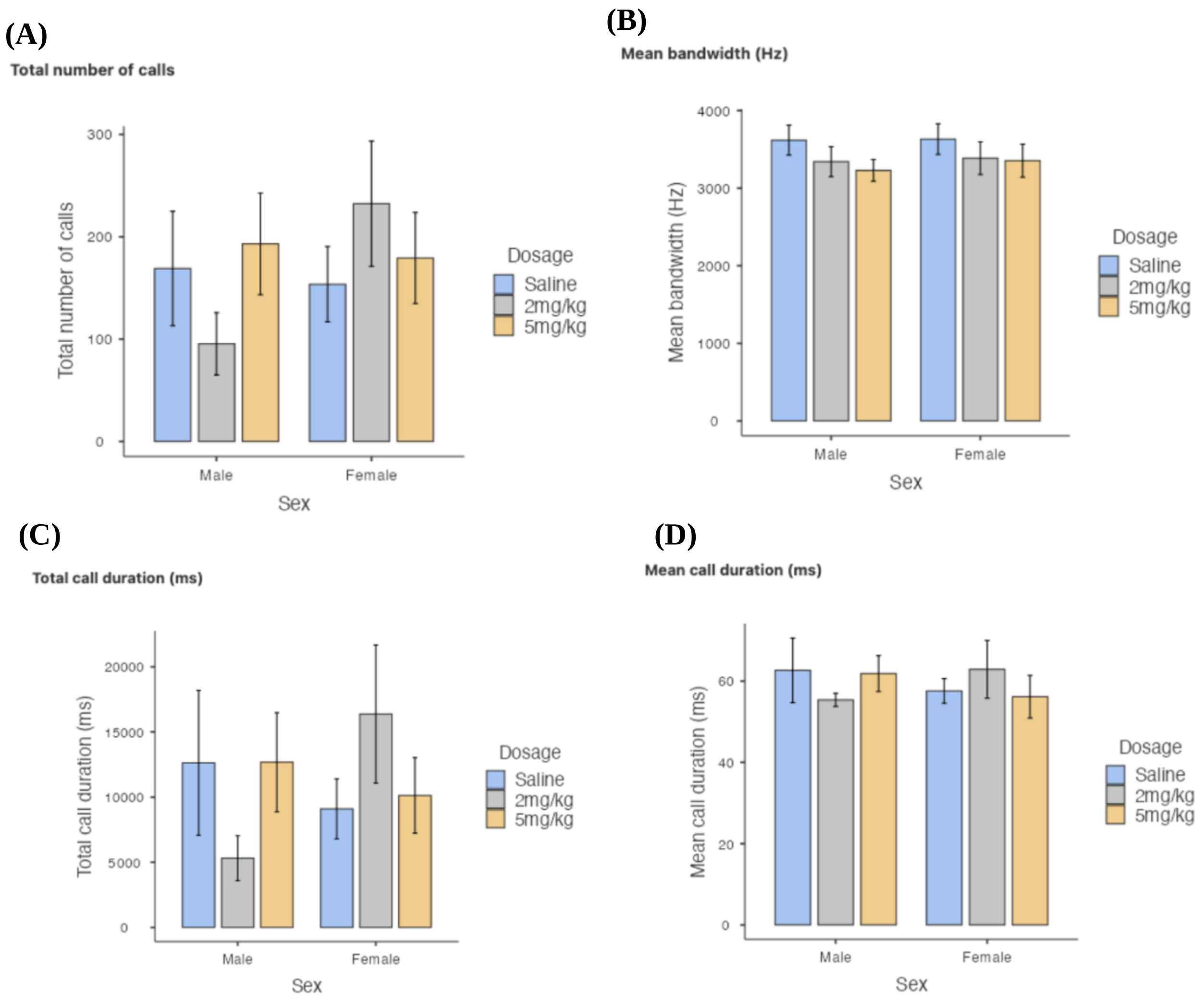


Fig. 3. **Acoustic characteristics of USVs in PND7 saline and poly(I:C) pups (full dataset; cohort 1 + 2).** Data are presented as the mean ± SD of (a) total number of calls, (b) average bandwidth, (c) total call duration and (d) mean call duration. Number of pups: saline: $N = 12$; 2mg/kg poly(I:C): $N = 11$; 5mg/kg poly(I:C): $N = 14$.

Effects of poly(I:C) on total number of USVs

No significant main effect for sex ($F(1, 31) = 0.81$, $p > 0.3$) or dosage ($F(2, 31) = 0.17$, $p > 0.8$) was found for the PND7 data. Despite males in the 2mg/kg poly(I:C) condition producing overall less calls compared to females in the 2mg/kg poly(I:C) condition, no significant interaction effect (sex*dosage) was found ($F(2, 31) = 1.5$, $p > 0.2$).

Effects of poly(I:C) on mean bandwidth of USVs

Mean bandwidth was not significantly different across sex ($F(1, 31) = 0.15$, $p > 0.7$) or dosage ($F(2, 31) = 1.69$, $p > 0.2$) for the PND7 data. Additionally, no significant interaction effect (sex*dosage) for mean bandwidth was observed ($F(2, 31) = 0.05$, $p > 0.9$).

Effects of poly(I:C) on total USV call duration

Total call duration was calculated as the sum of the duration of calls across the 8 minutes of recordings for each pup, and then averaged across conditions. No significant main effect of sex ($F(1, 31) = 0.27$, $p > 0.6$) or dosage ($F(2, 31) = 0.01$, $p > 0.9$) were observed across for the PND7 data. While males in the 2mg/kg poly(I:C) condition had a comparatively less total call duration than females in the 2mg/kg poly(I:C) condition which was nearing significance, no significant interaction effect (sex*dosage) was observed ($F(2, 31) = 2.01$, $p > 0.1$).

Effects of poly(I:C) on mean USV call duration

Mean call duration was calculated as the average length of the individual USVs. No significant main effect for sex ($F(1, 31) = 0.06$, $p > 0.8$) or dosage ($F(2, 31) = 0.03$, $p > 0.9$)

were observed for the PND7 mean call duration data. Additionally, no interaction effect (sex*dosage) was observed for mean call duration ($F(2,31) = 0.87$, $p > 0.4$).

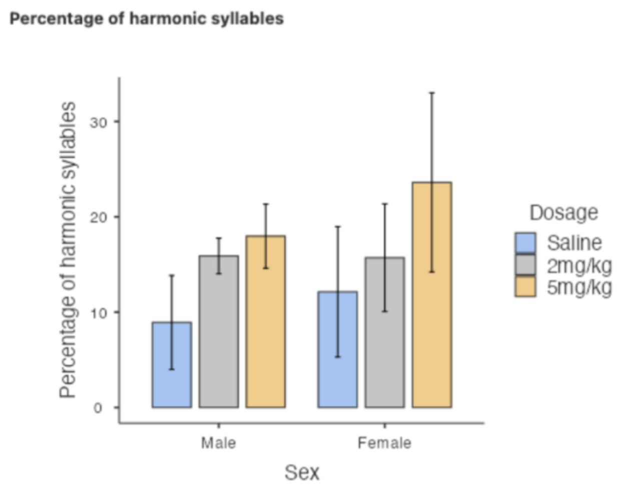


Fig. 4. **Percentage of harmonic syllables in PND7 saline and poly(I:C) pups (cohort 2).** Data are presented as the mean ± SD. Number of pups: saline: $N = 9$; 2mg/kg poly(I:C): $N = 7$; 5mg/kg poly(I:C): $N = 10$.

Effects of poly(I:C) on percentage of harmonic syllables

The analysis of harmonic syllables was conducted by calculating the number of harmonic calls as a percentage of the total number of calls for each pup, then averaged across conditions. This analysis was conducted separately only on the cohort 2 data. For the complete ANOVA table for percentage of harmonic syllables, see figure B1.

Additionally to the exclusion of the aforementioned pup (2mg/kg poly(I:C) male) from the entire dataset, one further male pup from the saline condition was excluded specifically from the analysis of harmonic syllables for having a total percentage of harmonic syllables of 72.7% of the total calls, which was over 3 standard deviations above the mean for the dataset ($M = 18.3$, $SD = 17.1$). For the full dataset including this outlier in the harmonic syllables analysis, see figure B2.

No significant main effect was found for sex ($F(1, 20) = 0.36$, $p > 0.5$). There does appear to be a trend for dosage, where pups in both the 2mg/kg poly(I:C) and 5mg/kg poly(I:C) treatment conditions seem to produce an overall greater percentage of harmonic syllables than saline controls, however, no significant main effect of dosage was observed ($F(2, 20) = 1.79$, $p > 0.1$). No significant interaction effect (sex*dosage) was observed ($F(2, 20) = 0.11$, $p > 0.9$).

Discussion

A robust link between maternal infection and neurodevelopmental disorders in offspring later in life, including ASD, SZ and mood disorders (Atladóttir et al., 2010; Khandaker et al., 2013; Al-Haddad et al., 2019) has been elucidated by a large corpus of studies. The present study explored the impact of prenatal infection, simulated through maternal exposure to the viral mimic poly(I:C), on multiple acoustic characteristics of offspring USVs; understood to be an early indicator of neurodevelopment in rat pups. In line with previous literature, we expected to observe alterations in offspring USVs departing from the typical developmental trajectory of USV production and investigated if any observed changes depended on dose or sex. No significant differences were found between the saline controls and the two poly(I:C) treatment conditions on the outcome parameters of total call number, mean call bandwidth, total call duration, mean call duration or percentage of harmonic syllables. No significant sex differences and no significant interaction effects between sex and dosage were observed on any of the investigated parameters.

Call rate did not significantly differ between poly(I:C) treatment condition pups and saline controls. This was unexpected, given the existing literature has consistently found prenatal poly(I:C) administration reduces the total number of isolation calls in offspring (Chou et al., 2015; Potasiewicz et al., 2020; Piotrowska et al., 2019). Moreover, considering the socio-affective functionality of these separation calls (Jouda et al., 2019) and that neurodevelopmental disorders feature deficits in social functioning and communication (Hommer & Swedo, 2015), failure to find any alterations in call rate perhaps challenges the face validity of these separation calls as an early indicator of neurodevelopmental abnormalities. However, given such consistent findings in the existing literature, the present null findings more plausibly reflect a small sample size. A trend emerged where males in the 2mg/kg poly(I:C) condition produced less overall calls compared to females in the 2mg/kg poly(I:C) condition. While not significant, this may reflect the higher prevalence of neurodevelopmental disorders in males. Additionally, this trend may corroborate the MSA hypothesis and the findings of Goeden et al. (2016), where a low dose of 2mg/kg poly(I:C) was sufficient to induce neurodevelopmental changes, not through an increase in proinflammatory cytokines, but through an increase in placental serotonin. Further research with a larger sample is required to investigate if a low dose of 2mg/kg poly(I:C) reliably alters USV call rate, which would advance the MSA hypothesis as a potential underlying mechanism. Our findings alone might also suggest that 5mg/kg poly(I:C) is too high of a dose to produce functional abnormalities. However, given the majority of the previous literature has found reduced call rate following a high dose of poly(I:C), between 4-5mg/kg (Chou et al., 2015; Potasiewicz et al., 2020; Piotrowska et al., 2019) this speculation is unlikely.

Of note, one methodological difference consistent across all previous studies that found a significant reduction in call rate was that poly(I:C) was administered on GD15 (Chou et al., 2015; Potasiewicz et al., 2020; Piotrowska et al., 2019). Conversely, poly(I:C) was administered on GD12 in the present study. Therefore, a more plausible account of the present null findings is that perhaps it is not the dosage, rather the timing of the infection that is more critical in ensuing neurodevelopmental abnormalities. Future research should compare different timings of prenatal poly(I:C) administration on functional outcomes in offspring to investigate this possibility.

In the present study, mean bandwidth did not significantly differ between poly(I:C) treatment groups and saline controls. Potasiewicz et al. (2020) reported no reductions in bandwidth following maternal poly(I:C) exposure, while Wilkin-Krug et al. (2022) found calls from poly(I:C) treated pups had reduced bandwidth. Notably, Potasiewicz et al. (2020) used 18-24 pups per condition, while Wilkin-Krug et al. (2022) used 32 pups per condition. It may be that a larger sample size is necessary to reveal the effects of poly(I:C) on bandwidth. Indeed, although insignificant, a trend emerged where the mean bandwidth of poly(I:C) treated offspring was slightly lower than saline controls - a trend which may have eventuated had we used a greater number of pups. Such findings suggest poly(I:C) lowers the complexity of separation calls, however, future research with a larger sample size is required to confirm this.

No significant differences in total call duration or mean call duration were found between poly(I:C) treatment conditions and saline controls. This finding aligns with previous literature that has consistently failed to identify an effect of poly(I:C) on call duration (Wilkin-Krug et al., 2022; Potasiewicz et al., 2020). Stark et al. (2020) report that stress caused by maternal separation affects call duration in healthy pups significantly less than call rate, indicating perhaps call duration is not a sensitive measure of early neurodevelopmental alterations, possibly accounting for the present null findings.

In the present study, prenatal poly(I:C) treatment did not significantly reduce the number of harmonic syllables compared to saline controls, despite previous literature reporting poly(I:C) administration reduces the number of harmonic syllables (Premoli et al., 2021; Malkova et al., 2012). One important consideration is that the aforementioned studies finding a reduction in the number of harmonics used mice. It is, therefore, possible that species differences may account for the discrepancy in our results. Considering harmonic syllables are speculated to index the emotionality of a call, and that neurodevelopmental disorders are characterised by deficits in emotional rapport (Hommer & Swedo, 2015), it is interesting that a trend emerged where poly(I:C) treated offspring produced a greater percentage of harmonic syllables than saline controls. This perhaps reflects a gap in our understanding of the meaning of these calls. Interestingly, harmonic syllables seemed to be qualitatively different across individual pups. Some would emit longer harmonics in more repetitive patterns, whereas others would emit shorter harmonics at more random intervals. We might speculate that perhaps it is not the rate of harmonic syllables, but rather the qualitative features or patterns of these syllables that might better index neurodevelopmental abnormalities; patterns and qualitative features lost through aggregating the data. As this is a relatively novel field, it is difficult to reach any conclusions, however future research might benefit from fully characterising the qualitative aspects of these calls to further our understanding of their specific meanings.

Finally, no significant differences between males and females were found for any of the outcome parameters. This finding was unexpected as males are disproportionately affected by prenatal poly(I:C) treatment within the exisiting literature (Potasiewicz et al., 2020; Piotrowska et al., 2019; Wilkin-Krug et al., 2022), which aligns with the higher prevalence of neurodevelopmental disorders observed in males (Han et al., 2021). As previously mentioned, a trend emerged where males in the 2mg/kg poly(I:C) condition had a lower call rate than females in the 2mg/kg poly(I:C) condition. Although insignificant, we might speculate this reflects the higher prevalence and severity of neurodevelopmental disorders observed in males, and substantiates evidence that the male brain has a more inflammatory environment than females during early development, possibly contributing to increased susceptibility to implications from maternal immune activation.

The present study was not without limitations. Foremost is the possibility that *baseline* USV emissions were not measured. Isolation, along with other stressful situations such as handling and temperature reductions, reliably affect USV emissions (Stark et al., 2020). Despite precautions taken to minimise stress (temperature controlled ECGenie holder, 15 minutes maximum separation), the isolation procedure remains a stressful experience, which likely confounded our baseline USV recordings. Additionally, Spratt et al. (2012) investigated cortisol levels in children with and without ASD in response to a stressful situation and found that children with ASD had a comparatively prolonged stress response and took longer to calm down. Considering poly(I:C) treatment evidently contributes to functional outcomes consistent with ASD, we might speculate that offspring of poly(I:C) treated dams similarly have a prolonged stress response compared to saline controls, which may have interfered with their baseline USV emissions and our between-group comparisons. It would be valuable for future studies to investigate how the acoustic parameters of USVs change under stressful conditions compared to baseline, or look to improve the isolation method to further minimise stress.

There were also some limitations within our methodology. Most pertinently was the use of the Raven Pro software to analyse USVs. The quality of the audio files impacted the detection accuracy of the software, which forced a manual analysis of the audio files, a method that is obviously prone to human error which has implications for the validity of the results. Additionally, the PND14 audio data was unviable. It is not known whether technical issues arose from a malfunction with the microphone, or if excessive movement and escaping of the ECGenie holder, likely due to the larger size of the PND14 pups, contributed to noise in the audio files. Nonetheless, not being able to compare USVs across postnatal days was a limitation of the study. Acoustic parameters of USVs reliably change with age, and such comparison may have provided invaluable data to better elucidate the effects of prenatal poly(I:C) treatment, more so than extrapolating from a single day.

Despite no significant results observed across any of the outcome parameters, the emerging trends within the data seem to suggest prenatal poly(I:C) treatment, particularly at a low dose of 2mg/kg and specifically in males, may contribute to functional outcomes consistent with neurodevelopmental abnormalities, as indexed by changes in USV emissions. Without a larger sample size, these possibilities are limited to speculation. Notwithstanding, we cannot discount the potential relevance of these trends for the current COVID-19 pandemic, as they corroborate the vulnerability of the children of pregnant mothers who have contracted this virus, and potentially women who received a vaccine whilst pregnant, as vaccines are known to stimulate a brief immune challenge. Given these implications, it is imperative to further elucidate the consequences of prenatal infection on neurodevelopmental disorders with a larger sample size, to better understand the role of severity of infection. Additionally, our preliminary data suggest research into other potential factors underpinning the relationship between prenatal infection and neurodevelopmental abnormalities, particularly the timing of the infection, may be a valuable future avenue.

References

Al-Haddad, B. J. S., Jacobsson, B., Chabra, S., Modzelewska, D., Olson, E. M., Bernier, R., Enquobahrie, D. A., Hagberg, H., Östling, S., Rajagopal, L., Adams Waldorf, K. M., & Sengpiel, V. (2019). Long-term risk of neuropsychiatric disease after exposure to infection in utero. *JAMA Psychiatry*, *76*(6), 594–602.

Atladóttir, H. Ó., Thorsen, P., Østergaard, L., Schendel, D. E., Lemcke, S., Abdallah, M., & Parner, E. T. (2010). Maternal infection requiring hospitalization during pregnancy and autism spectrum disorders. *Journal of Autism and Developmental Disorders*, *40*(12), 1423–1430.

Boksa, P. (2008). Maternal infection during pregnancy and schizophrenia. *Journal of Psychiatry & Neuroscience*, *33*(3).

Brudzynski, S. M., Kehoe, P., & Callahan, M. (1999). Sonographic Structure of Isolation-Induced Ultrasonic Calls of Rat Pups. *Developmental Psychobiology, 34*.

Caruso, A., Ricceri, L., & Scattoni, M. L. (2020). Ultrasonic vocalizations as a fundamental tool for early and adult behavioral phenotyping of Autism Spectrum Disorder rodent models. *Neuroscience and Biobehavioral Reviews*, *116*, 31–43.

Chou, S., Jones, S., & Li, M. (2015). Adolescent olanzapine sensitization is correlated with hippocampal stem cell proliferation in a maternal immune activation rat model of schizophrenia. *Brain Research*, *1618*, 122–135.

Cox, E. T., Hodge, C. W., Sheikh, M. J., Abramowitz, A. C., Jones, G. F., Jamieson-Drake, A. W., Makam, P. R., Zeskind, P. S., & Johns, J. M. (2012). Delayed developmental changes in neonatal vocalizations correlates with variations in ventral medial hypothalamus and central amygdala development in the rodent infant: Effects of prenatal cocaine. *Behavioural Brain Research*, *235*(2), 166–175.

Fish, E. W., Faccidomo, S., Gupta, S., & Miczek, K. A. (2004). Anxiolytic-Like Effects of Escitalopram, Citalopram, and R-Citalopram in Maternally Separated Mouse Pups. *Journal of Pharmacology and Experimental Therapeutics*, *308*(2), 474–480.

Goeden, N., Velasquez, J., Arnold, K. A., Chan, Y., Lund, B. T., Anderson, G. M., & Bonnin, A. (2016). Maternal inflammation disrupts fetal neurodevelopment via increased placental output of serotonin to the fetal brain. *Journal of Neuroscience*, *36*(22), 6041–6049.

Han, V. X., Patel, S., Jones, H. F., & Dale, R. C. (2021). Maternal immune activation and neuroinflammation in human neurodevelopmental disorders. *Nature Reviews Neurology*, *17*(9), 564–579.

Harrington, R. A., Lee, L. C., Crum, R. M., Zimmerman, A. W., & Hertz-Picciotto, I. (2013). Serotonin Hypothesis of Autism: Implications for Selective Serotonin Reuptake Inhibitor Use during Pregnancy. *Autism Research*, *6*(3), 149–168.

Hodgson, R. A., Guthrie, D. H., & Varty, G. B. (2008). Duration of ultrasonic vocalizations in the isolated rat pup as a behavioral measure: Sensitivity to anxiolytic and antidepressant drugs. *Pharmacology Biochemistry and Behavior*, *88*(3), 341–348.

Hommer, R. E., & Swedo, S. E. (2015). Schizophrenia and autism-related disorders. *Schizophrenia Bulletin*, *41*(2), 313–314.

Jouda, J., Wöhr, M., & del Rey, A. (2019). Immunity and ultrasonic vocalization in rodents. *Annals of the New York Academy of Sciences*, *1437*(1), 68–82.

Khandaker, G. M., Zimbron, J., Lewis, G., & Jones, P. B. (2013). Prenatal maternal infection, neurodevelopment and adult schizophrenia: A systematic review of population-based studies. *Psychological Medicine*, *43*(2), 239–257.

Liu, D., Diorio, J., Tannenbaum, B., Caldji, C., Francis, D., Freedman, A., Sharma, S., Pearson, D., Plotsky, P. M., & Meaney, M. J. (1997). Maternal care, hippocampal glucocorticoid receptors, and hypothalamic-pituitary-adrenal responses to stress. *Science*, *277*, 1659–1661.

Malkova, N. v., Yu, C. Z., Hsiao, E. Y., Moore, M. J., & Patterson, P. H. (2012). Maternal immune activation yields offspring displaying mouse versions of the three core symptoms of autism. *Brain, Behavior, and Immunity*, *26*(4), 607–616.

Mednick, S. A., Machon, R. A., Huttunen, M. O., & Bonett, D. (1988). *Adult Schizophrenia Following Prenatal Exposure to an Influenza Epidemic*.

Patterson, P. H. (2011). Maternal infection and immune involvement in autism. *Trends in Molecular Medicine*, *17*(7), 389–394.

Pendyala, G., Chou, S., Jung, Y., Coiro, P., Spartz, E., Padmashri, R., Li, M., & Dunaevsky, A. (2017). Maternal Immune Activation Causes Behavioral Impairments and Altered Cerebellar Cytokine and Synaptic Protein Expression. *Neuropsychopharmacology*, *42*(7), 1435–1446.

Piotrowska, D., Sopel, J., Potasiewicz, A., Popik, P., & Nikiforuk, A. (2019). Pups' altered ultrasonic vocalisation in the poly I:C rat model of autism spectrum disorder: results from the mother isolation test. *Acta Neurobiologiae Experimentalis*, *79*.

Porter, F. L., Miller, R. H., & Marshall, R. E. (1986). Neonatal Pain Cries: Effect of Circumcision on Acoustic Features and Perceived Urgency. *Child Development*, *57*(3).

Potasiewicz, A., Gzielo, K., Popik, P., & Nikiforuk, A. (2020). Effects of prenatal exposure to valproic acid or poly(I:C) on ultrasonic vocalizations in rat pups: The role of social cues. *Physiology and Behavior*, *225*.

Premoli, M., Memo, M., & Bonini, S. (2021). Ultrasonic vocalizations in mice: Relevance for ethologic and neurodevelopmental disorders studies. *Neural Regeneration Research*, *16*(6), 1158–1167.

Prins, J. R., Eskandar, S., Eggen, B. J. L., & Scherjon, S. A. (2018). Microglia, the missing link in maternal immune activation and fetal neurodevelopment; and a possible link in preeclampsia and disturbed neurodevelopment? *Journal of Reproductive Immunology, 126*, 18–22.

Robbins, J. R., & Bakardjiev, A. I. (2012). Pathogens and the placental fortress. *Current Opinion in Microbiology*, *15*(1), 36–43.

Spratt, E. G., Nicholas, J. S., Brady, K. T., Carpenter, L. A., Hatcher, C. R., Meekins, K. A., Furlanetto, R. W., & Charles, J. M. (2012). Enhanced cortisol response to stress in children in autism. *Journal of Autism and Developmental Disorders*, *42*(1), 75–81.

Stark, R. A., Harker, A., Salamanca, S., Pellis, S. M., Li, F., & Gibb, R. L. (2020). Development of ultrasonic calls in rat pups follows similar patterns regardless of isolation distress. *Developmental Psychobiology*, *62*(5), 617–630.

Tioleco, N., Silberman, A. E., Stratigos, K., Banerjee-Basu, S., Spann, M. N., Whitaker, A. H., & Turner, J. B. (2021). Prenatal maternal infection and risk for autism in offspring: A meta-analysis. *Autism Research*, *14*(6), 1296–1316.

Wilkin-Krug, L. C. M., Macaskill, A. C., & Ellenbroek, B. A. (2022). Preweaning environmental enrichment alters neonatal ultrasonic vocalisations in a rat model for prenatal infections. *Behavioural Pharmacology, 33*(6), 402–417.

Wöhr, M., & Schwarting, R. K. W. (2008). Maternal Care, Isolation-Induced Infant Ultrasonic Calling, and Their Relations to Adult Anxiety-Related Behavior in the Rat. *Behavioral Neuroscience*, *122*(2), 310–33

Appendix A

Complete ANOVA tables for combined cohort 1 and cohort 2 data for all measured dependent variables.

ANOVA - Total number of calls

	Sum of Squares	df	Mean Square	F	p
Sex	11808	1	11808	0.809	0.375
Dosage	4878	2	2439	0.167	0.847
Sex ✻ Dosage	44039	2	22019	1.509	0.237
Residuals	452251	31	14589		

Fig. A1. Complete ANOVA table for total number of calls for cohort 1 and cohort 2 combined data

ANOVA - Mean bandwidth (Hz)

	Sum of Squares	df	Mean Square	F	p
Sex	34405	1	34405	0.1509	0.700
Dosage	771123	2	385561	1.6909	0.201
Sex ✻ Dosage	21808	2	10904	0.0478	0.953
Residuals	7.07e+6	31	228017		

Fig. A2. Complete ANOVA table for mean bandwidth for cohort 1 and cohort 2 combined data

ANOVA - Total call duration (ms)

	Sum of Squares	df	Mean Square	F	p
Sex	2.52e+7	1	2.52e+7	0.2650	0.610
Dosage	2.61e+6	2	1.30e+6	0.0137	0.986
Sex ✻ Dosage	3.82e+8	2	1.91e+8	2.0065	0.152
Residuals	2.95e+9	31	9.51e+7		

Fig. A3. Complete ANOVA table for total call duration for cohort 1 and cohort 2 combined data

ANOVA - Mean call duration (ms)

	Sum of Squares	df	Mean Square	F	p
Sex	10.67	1	10.67	0.0579	0.811
Dosage	9.10	2	4.55	0.0247	0.976
Sex ✻ Dosage	320.76	2	160.38	0.8702	0.429
Residuals	5713.23	31	184.30		

Fig A4. Complete ANOVA table for mean call duration for cohort 1 and cohort 2 combined data

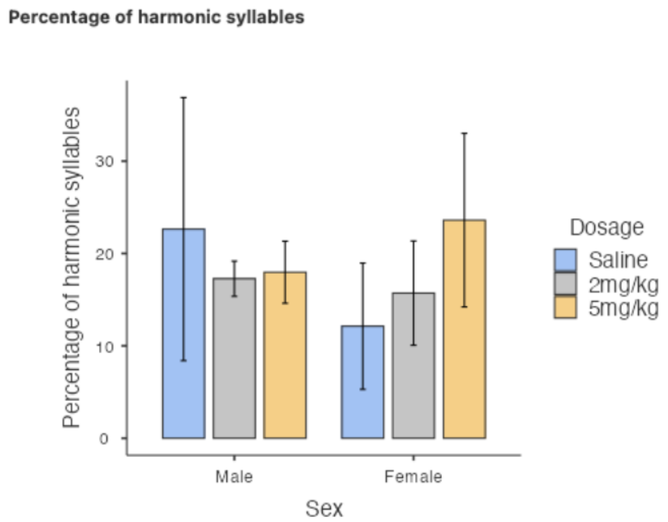
Appendix B

Complete ANOVA table for the percentage of harmonic syllables analysed from the cohort 2 data (excluding the outlier), as well as the graphed results and ANOVA table for the percentage of harmonic syllables which includes the outlier that was 3SD above the mean.

ANOVA - Percentage of harmonic syllables

	Sum of Squares	df	Mean Square	F	p
Sex	64.3	1	64.3	0.355	0.558
Dosage	649.8	2	324.9	1.796	0.192
Sex ∗ Dosage	38.1	2	19.0	0.105	0.901
Residuals	3619.1	20	181.0		

Fig. B1. Complete ANOVA table for percentage of harmonic syllables for cohort 2 data which excludes the outlier



ANOVA - Percentage of harmonic syllables

	Sum of Squares	df	Mean Square	F	p
Sex	31.8	1	31.8	0.0944	0.762
Dosage	96.5	2	48.3	0.1434	0.867
Sex ∗ Dosage	327.0	2	163.5	0.4859	0.622
Residuals	7404.3	22	336.6		

Fig. B2. Graph showing the percentage of harmonic syllables for cohort 2 which includes the outlier, as well as the complete ANOVA table for the percentage of harmonic syllables for cohort 2 which includes the outlier.